Set-up: $\mathcal{O}_0'$ regular block of category $\mathcal{O}$

$\lambda = 0$ highest weight

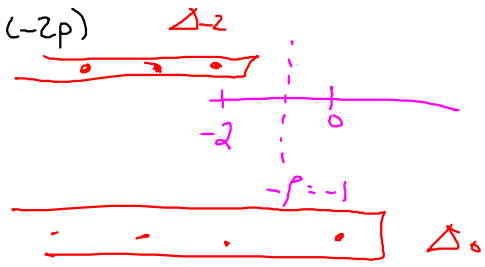
$\lambda = -2\rho$ lowest weight.

$\nabla_w = \nabla_{w\cdot(-2\rho)}$ antidom

$\Delta_w = \Delta_{w\cdot(-2\rho)}$

as w longer ∇_w, Δ_w bigger

$w \in W$



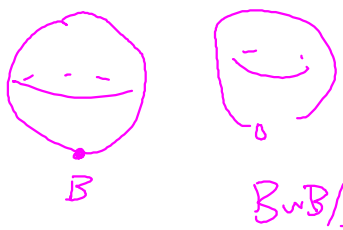
D-modules:

$X = G/B$

$X_w = BwB/B$ · B-orbits on X

"Schubert stratification"

e.g. $G = SL_2$



ie w longer, $\dim(X_w)$ grows.

$DMod^N(G/B)$

$\cong$

$DMod^B(G/N)$

$i_w : X_w \hookrightarrow X$

Goal: Know $\Gamma(G/B, -) : DMod(N\backslash G/B) \underset{\simeq}{\overset{\simeq}{\rightleftarrows}} \mathcal{O}_0' : Loc(-)$

$\Gamma(G/B, i_{w*}\mathcal{O}_{X_w}) \simeq \nabla_w$

$\Gamma(G/B, i_{w!}\mathcal{O}_{X_w}) \simeq \Delta_w$

$\Rightarrow \Gamma(G/B, L(X_w, \mathcal{O}_{X_w})) \simeq L_w$ ← complete list of simples on LHS (wrt Sch. stratification)

Remark: X_w affine spaces $\simeq \mathbb{A}^m$

$\Rightarrow$ only on local system

Strategy: ① Characterize Δ_w, ∇_w by · categorical properties
· characters $= \sum_{\alpha \in \Phi^+} \dim(M_\alpha) e^{-\alpha}$

② Check that $\Gamma(G/B, -)$ has these properties.

③ "Get" characters, need to study geometry of Schubert strat.

Lemma: Suppose $M \in \mathcal{O}_0'$, ∇_w

· have the same composition factors

· $\mathrm{Hom}(L_{w'}, M) = 0$ for $\ell(w') < \ell(w)$

Then $M \cong \nabla_w$.

$\Uparrow$

$\Downarrow$

$M \in \mathcal{O}_0^1$ • M, Δ_w have same c. factors.

• $\mathrm{Hom}(M, L_{w'}) = 0$ for $\ell(w') < \ell(w)$

$\Rightarrow M \cong \Delta_w$.

Proof: • have the same character (i.e. dim of $\mathfrak{h}$-eigenspaces)

$\Rightarrow$ M has highest weight $w \cdot (-2\rho)$

$\Rightarrow$ nonzero map $\phi : \Delta_w \to M$

surjective $M \twoheadrightarrow \mathrm{coker}(\phi) \to L_{w'}$ nonzero. must $\ell(w') < \ell(w)$

$\Rightarrow w = w'$

So, L_w in $\mathrm{coker}(\phi)$ $\Rightarrow L_w$ in $\ker(\phi)$ contradiction

So ϕ must have been surjective, so ... iso (i.e. same composition factors). $\square$

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ch: $K_0(\mathcal{O}_0^1) \to \mathbb{Z}[X^*(H)]$ fm

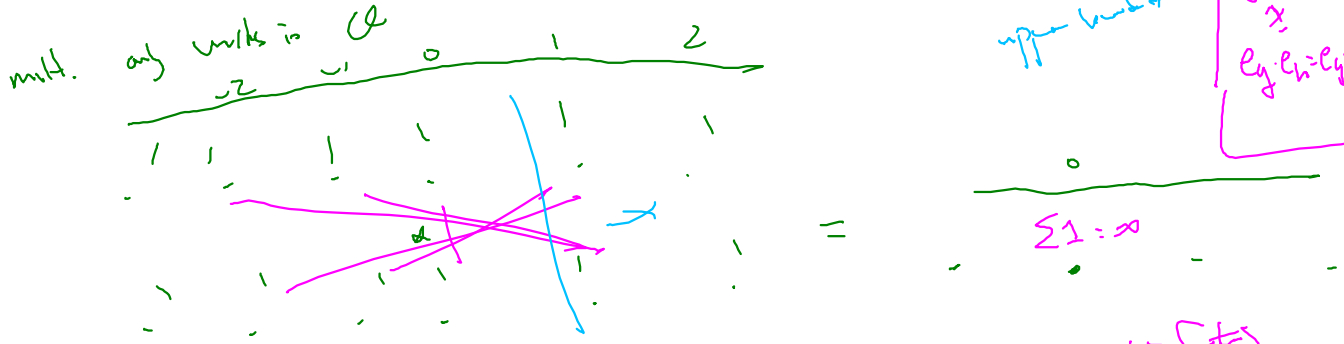
$\mathrm{ch}(M) = \sum \dim(M_\lambda) e^\lambda$ weight space

Note: addition $e^\lambda e^\mu = e^{\lambda+\mu}$

multiplication commutative

well defined upper triangular

mult. only works in $\mathbb{Q}$



G discrete

$e_g \cdot e_h = 0$ $k[G]$

$\neq$ $\cong$

$e_g \cdot e_h = e_{gh}$ kG

Prop: ch is injective.

Update: ch agree $\Leftrightarrow$ composition factors agree

Sketch: $[M_w]$ $[L_w]$ these are basis, i.e. linearly independent

+ K_0

Thm: $\Gamma(G/B, i_{w*}\mathcal{O}_{X_w}) \simeq \Delta_w$

$\Gamma(G/B, i_{w!}\mathcal{O}_{X_w}) \simeq \nabla_w$.

geometry

Pf: ① Again then, assuming we can check that $\cdot \hookrightarrow G/B$ characters agree.

Induction on $\ell(w)$. Base case: $\Gamma(G/B, i_!\mathcal{O}_{X_e}) = i_*\mathcal{O}_{X_e}$ ✓

has right character.

costalks

Induct: $\mathrm{Hom}(L_{w'}, \Gamma(G/B, i_{w*}\mathcal{O}_{X_w}))$

by induction, $L_{w'}$ such if $i_{w'*}\mathcal{O}_{X_{w'}}$ $\Rightarrow$ supported on $\overline{X_{w'}} = \bigcup_{w \le w'} X_{w''}$

$\longrightarrow \mathrm{Hom}(L_{w'}, \Gamma(G/B, i_{w*}\mathcal{O}_{X_w}))$

$= \mathrm{Hom}(\mathrm{Loc}(L_{w'}), i_{w*}\mathcal{O}_{X_w})$

$= \mathrm{Hom}(i_w^* \mathrm{Loc}(L_{w'}), \mathcal{O}_{X_w}) = \mathrm{Hom}(0, \mathcal{O}_{X_w})$

$= 0$.

$X_w \cap \mathrm{supp}(L_{w'})$

$= X_w \cap \overline{X_{w'}} = \phi$

② Compute characters:

$U \subseteq B$ unipotent

$U^- \subseteq B^-$ opposite.

$B \cap B^- = T$

$U_w B = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} B$

$\begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix} B$

act trivially on X_w

$\boxed{wU^-B/B}$

$X_w = BwB/B = U_w B/B = U_w^+ wB/B \subseteq Y_w = U_w^- U_w B/B \hookrightarrow X$

$X_w \hookrightarrow T$ fixed points

Y_w open in G/B. (not union of strata)

contain all normal directions to strata

$\subseteq$ cl. $\cap$)

$U_w^- = e$

Sl2: $UB \longrightarrow U_w^- UB$

$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \longrightarrow$

$\begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix} \longrightarrow$

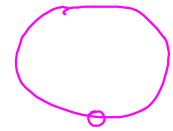
$U_w UB$

$U_w^- U_w UB$

Sk: $\boxed{U^-B/B}$ $\quad U^- = \begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix}$ $\quad x = 0$ $\quad \xrightarrow{wU^-B}$

$\overset{\cup}{B} = \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix}$

B/B

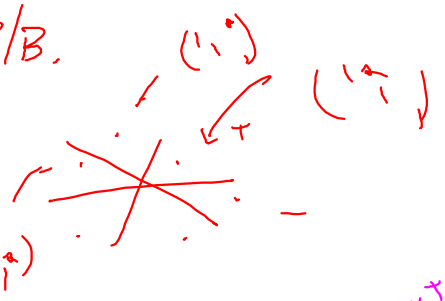


U^-B/B open affine of G/B contains B/B (T-fixed point)

wU^-B/B contains wB/B. other T-fixed pts. points

What is U_w^+? $\quad X_w = BwB/B = U_w^+ \, wB/B$.

- Notation α root, $U_\alpha \subseteq G$ $\cong \mathbb{G}_a$



Fact: $U^+ = \prod_{\alpha \in \Delta^+} U_\alpha \qquad U^- = \prod_{\alpha \in \Delta^-} U_\alpha$.

- $U_\alpha wB/B = wU_{w\cdot\alpha}B/B \overset{?}{=} wB/B$ (stabilizer?) $\longrightarrow$ when $\alpha \in \Delta^+$, $w\cdot\alpha \in \Delta^+$, $\alpha \in \Delta^+ \cap w\cdot(\Delta^+)$

remaining stuff generate orbit

$\alpha \in \Delta^+$, $w\cdot\alpha \notin \Delta^+$ i.e. $w\cdot\alpha \in \Delta^-$

$\boxed{\alpha \in \Delta^+ \cap w\cdot(\Delta^-)}$

$\alpha \in \Delta^- \cap w\cdot(\Delta^+) \xleftarrow{\text{Jordan}} U_w^+ \hookleftarrow \boxed{\alpha \in \Delta^+ \cap w\cdot(\Delta^-)}$

normal. $U_w^- \hookleftarrow \alpha \in \Delta^- \cap w\cdot(\Delta^-)$

union.

$\Delta^- \quad \mathrm{Sym}(U^-) \leftarrow$ character of Δ_0